A

Project Report on

SafeSurf — A Machine Learning-Based Real-Time Phishing Website Detector

Submitted in partial fulfillment of the requirements
for the award of the degree of

Bachelor of Technology
in
Computer Science and Engineering

Submitted by

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Under the esteemed supervision of

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(Approved by AICTE, New Delhi & Affiliated to JNTUK, Kakinada)
Surampalem, East Godavari District
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2022–2026

ADITYA COLLEGE OF ENGINEERING & TECHNOLOGY (A)

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Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the project work entitled, **"SafeSurf — A Machine Learning-Based Real-Time Phishing Website Detector"**, is a bonafide work carried out by **YALLAPRAGADA OMTEJA (22P31A0539)**, **KARRI DEEPIKA (22P31A0517)**, **DANGETI SAI GAYATRI VISHWESH (22P31A0508)**, **ANKIT ROY CHOWDHURY (22P31A0504)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** from Aditya College of Engineering and Technology, Surampalem, during the academic year 2025–2026.

This project work has not been submitted in full or part to any other University or educational institute for the award of any degree or diploma.

PROJECT SUPERVISOR	HEAD OF THE DEPARTMENT
Mr. Alla Rajendra	**Dr. CHAKKA S.V.S.N. Murty**
Assistant Professor	**Professor**
M.Tech, (Ph.D)	M.Tech, Ph.D

EXTERNAL EXAMINER

DECLARATION

We hereby declare that this project entitled **"SafeSurf — A Machine Learning-Based Real-Time Phishing Website Detector"** has been undertaken by us and this work has been submitted to Department of Computer Science & Engineering, **ADITYA COLLEGE OF ENGINEERING AND TECHNOLOGY (A)**, Surampalem affiliated to JNTUK, Kakinada, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**.

We further declare that this project work has not been submitted in full or part to any other University or educational institute for the award of any degree or diploma.

PROJECT ASSOCIATES

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We feel elated to thank Dr. Pullela S.V.S.S. Ravi Kumar, Dean (Academics) of Aditya College of Engineering and Technology (A) for his cooperation in completion of our project.

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We avail this opportunity to express our deep sense and heartfelt thanks to the Management of Aditya College of Engineering & Technology (A) for providing great support for us by arranging the trainees, and facilities needed to complete our project and for giving us the opportunity for doing this work.

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VISION

To induce higher planes of learning by imparting technical education with

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- ✓ Value based instruction and to emerge as a premiere institute

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Achieving academic excellence by providing globally acceptable technicaleducation by forecasting technology through

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VISION:

To become a center for excellence in Computer Science and Engineering education and innovation.

MISSION:

- Provide state of art infrastructure
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- Organize socio cultural events for better society
- Undertake collaborative works with academia and industry
- Encourage students and staff self-motivated, problem-solving individuals using Artificial Intelligence
- Encourage entrepreneurship in young minds.

Head of the Department

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROGRAM EDUCATIONAL OBJECTIVES

PEO1: Capability to design and develop new software products as per requirements of the various domains and eligible to take the roles in various government, research organizations and industry

PEO2: More enthusiastic to adopt new technologies and to improve existing solutions by reducing complexity which serves society requirements as per timeline changes

PEO3: With good hands-on basic knowledge and ready improve academic qualifications in India or Abroad

PEO4: Ability to build and lead the team to achieve organization goals.

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROGRAM SPECIFIC OUTCOMES

PSO 1: The ability to design and develop computer programs for analyzing the data.

PSO 2: The ability to analyze data & develop Innovative ideas and provide solution by adopting emerging technologies for real time problems of software industry.

PSO 3: To encourage the research in software field that contribute to enhance the standards of human life style and maintain ethical values.

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROGRAM OUTCOMES

1. ENGINEERING KNOWLEDGE: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. PROBLEM ANALYSIS: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. DESIGN/DEVELOPMENT OF SOLUTIONS: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. CONDUCT INVESTIGATIONS OF COMPLEX PROBLEMS: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. MODERN TOOL USAGE: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

6. THE ENGINEER AND SOCIETY: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.

7. ENVIRONMENT AND SUSTAINABILITY: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. ETHICS: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. INDIVIDUAL AND TEAM WORK: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. COMMUNICATION: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, give and receive clear instructions.

11. PROJECT MANAGEMENT AND FINANCE: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. LIFE-LONG LEARNING: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Head of the Department

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ABSTRACT

SafeSurf — A Machine Learning-Based Real-Time Phishing Website Detector

Phishing attacks remain a significant cybersecurity threat, targeting users through deceptive websites that mimic legitimate platforms to steal sensitive information. Rapid and accurate detection of such malicious websites is therefore essential for ensuring safe online interactions. This paper presents **SafeSurf**, a real-time phishing website detection system based on supervised machine learning techniques. The system analyzes multiple URL and website characteristics—including domain attributes, HTTPS usage, URL length, and structural HTML indicators—to effectively distinguish between phishing and legitimate websites.

Unlike traditional blacklist-based approaches, which often fail to detect newly generated phishing domains, **SafeSurf** employs data-driven learning methods trained on a dataset of phishing and benign URLs. The system incorporates preprocessing techniques such as handling missing values, feature normalization, and noise reduction to improve model reliability. Feature importance analysis is used to identify key indicators associated with phishing activity, while an ensemble-based machine learning approach combines multiple classification models to improve detection performance and robustness.

Experimental results show that the proposed system achieves high detection accuracy with efficient prediction time, making it suitable for real-time applications. By integrating machine learning with automated feature analysis, **SafeSurf** provides a scalable and adaptive solution for phishing detection and helps strengthen cybersecurity by reducing users' exposure to malicious websites.

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Guide Signature

1. INTRODUCTION

1.1 Purpose of the Project

Our project, **SafeSurf**, aims to develop a real-time phishing website detection system using advanced machine learning techniques. The system analyzes URL and website infrastructure characteristics to identify and classify malicious phishing websites before users interact with them. By combining automated feature extraction with a multi-model ensemble learning approach, **SafeSurf** provides accurate and reliable phishing detection for modern web environments.

The primary goal of the project is to enhance cybersecurity by enabling proactive detection of phishing attacks that attempt to steal sensitive user information such as login credentials, financial data, and personal details. The system is designed for internet users, organizations, and cybersecurity platforms to improve online safety and reduce exposure to malicious websites. Key benefits of **SafeSurf** include real-time detection, high classification accuracy, minimal user interaction, and automated threat analysis.

1.2 Scope of the Project

The scope of **SafeSurf** involves designing and implementing an intelligent phishing detection framework capable of analyzing URLs and website infrastructure data in real time. The system integrates machine learning models, infrastructure analysis, and browser-based detection mechanisms to provide effective protection against phishing threats. The system consists of the following components:

- **Data Collection** – Aggregation of phishing and legitimate URL datasets from public sources for training and evaluation.
- **Feature Extraction** – Analysis of domain and infrastructure characteristics such as DNS records, SSL certificate information, domain registration details, URL structure, and HTML patterns.
- **Machine Learning Model** – Implementation of a VotingClassifier ensemble combining multiple models including Logistic Regression, LinearSVC, Random Forest,

HistGradientBoosting, and Calibrated XGBoost to improve prediction accuracy.

- **Detection Architecture** – A layered detection pipeline where a DNS Guard module filters invalid domains, followed by asynchronous infrastructure analysis using DNS, SSL, and RDAP information extraction.
- **User Interface and Browser Extension** – Development of a web-based dashboard and Chrome extension that intercept URLs and provide users with a clear **Safe** or **Phishing** verdict in real time.

The system is designed to support applications in **browser security tools, enterprise cybersecurity systems, and web protection platforms**. Future enhancements may include large-scale cloud deployment, integration with threat intelligence feeds, and automated browser protection mechanisms.

1.3 Current System

Existing phishing detection systems primarily rely on blacklist-based approaches, rule-based filters, or simple machine learning models to identify malicious websites. While these systems provide a basic level of protection, they often struggle to detect newly generated phishing websites and rapidly evolving attack techniques. Many current approaches:

- Depend heavily on static blacklists that require continuous manual updates.
- Have limited capability to detect zero-day phishing domains that have not yet been reported.
- Use single-model classifiers, which may reduce detection accuracy and robustness.
- Lack real-time infrastructure analysis, preventing deeper inspection of domain behavior and hosting characteristics.

As phishing techniques become more sophisticated, traditional detection mechanisms are often insufficient to provide reliable protection against emerging threats.

1.4 Proposed System

The proposed system, **SafeSurf**, introduces an advanced phishing detection framework that combines infrastructure analysis with a machine learning ensemble approach to improve detection accuracy and reliability.

The system implements a layered detection architecture beginning with a deterministic DNS Guard module that filters invalid domain lookups. This is followed by concurrent infrastructure data extraction using asynchronous processing to analyze DNS records, SSL certificates, and RDAP domain registration information.

SafeSurf employs a highly optimized VotingClassifier ensemble model, integrating multiple machine learning algorithms including Logistic Regression, LinearSVC, Random Forest, HistGradientBoosting, and Calibrated XGBoost. By combining predictions from multiple classifiers, the system improves both precision and recall when identifying phishing websites, including previously unseen or zero-day threats.

To provide practical usability, **SafeSurf** includes a web-based dashboard and Chrome browser extension that intercept URLs and automatically displays a clear detection verdict—either **Safe** or **Phishing**. This design ensures minimal user interaction while maintaining real-time protection.

The system is designed with key objectives including real-time detection, scalability, high precision, and minimal user intervention. In the future, **SafeSurf** can be extended with cloud-based deployment, integration with global threat intelligence sources, and automated browser-level security mechanisms, enabling more robust protection against evolving phishing attacks.

2. REQUIREMENT ANALYSIS

2.1 Functional Requirements

Functional requirements define the core operations and capabilities of the **SafeSurf** system. These requirements describe how the system accepts input URLs, processes them using machine learning techniques, and provides phishing detection results. They ensure that the system performs accurate, real-time classification and meets user expectations.

2.1.1 Importance of Functional Requirements

Functional requirements act as the foundation of the **SafeSurf** system, guiding the design of the frontend, backend, and machine learning pipeline. They ensure that:

- The system performs accurate phishing detection
- The processing workflow is clearly defined
- The system meets real-time performance requirements
- Developers and testers clearly understand system behavior

2.1.2 Characteristics of Functional Requirements

- Define specific functionalities required for phishing detection
- Describe input (URL), processing (feature extraction and machine learning), and output (classification)
- Specify system rules such as threshold-based decision logic
- Focus on both user interaction and automated detection mechanisms
- Can be validated and verified using test cases

2.1.3 Core Functional Requirements of SafeSurf

2.1.3.1 User Interface & Interaction

- The system provides:
  - A React-based web application
  - A Chrome extension for real-time URL monitoring
- Users can:
  - Enter a URL manually through the web interface
  - Automatically analyze URLs through the browser extension
- The system displays results as — **Safe** or **Phishing Risk**
- The UI is designed to be:
  - Simple and intuitive
  - Responsive and user-friendly
  - Free from complex technical details (no raw probability display)

2.1.3.2 Machine Learning & AI Functionality

- The system uses a hybrid ensemble machine learning approach for phishing detection.
- Models used include:
  - Logistic Regression
  - Linear SVM
  - Random Forest
  - HistGradientBoosting
  - Calibrated XGBoost
- Additionally, A separate NLP-based Logistic Regression model processes textual features.
- Hyperparameters are tuned using an incremental hyperparameter tuning approach.

2.1.3.3 URL Processing & Feature Extraction

- The system accepts a URL as input.
- Performs preprocessing including:
  - URL normalization (canonicalization)
  - Removal of unnecessary characters
- Feature extraction includes.
  - Lexical features (URL structure, length, symbols)
  - Network-based features (WHOIS, SSL, domain age)
  - NLP features (Bag-of-Words representation)
- Feature extraction is performed using Parallel and asynchronous processing.

2.1.3.4 Security & Validation

- The system includes a DNS Guard Layer that:
  - Detects invalid or non-existent domains (NXDOMAIN)
  - Blocks such requests before ML processing
- Input validation ensures:
  - Proper URL format
  - Handling of invalid or malformed inputs
- Prevents unnecessary computation and improves system efficiency

2.1.3.5 Real-Time Detection & Response

- The system processes requests in real time
- Uses a FastAPI backend for high-speed API handling
- Output includes:
  - Phishing classification result
  - Risk level (High / Medium / Safe)
  - JSON response for frontend and extension

2.1.3.6 Performance & Scalability

- Supports multiple concurrent users using:
  - Concurrency control mechanisms
- Optimized for:
  - Low latency predictions
  - Efficient resource usage
- Can be deployed on:
  - Cloud platforms (AWS, Google Cloud)

2.1.3.7 Reliability & Maintainability

- Ensures consistent and accurate predictions
- Modular architecture allows:
  - Easy updates to ML models
  - Extension of features
  - Backend improvements
- Supports:
  - Testing and debugging
  - Version control using Git

2.1.3.8 System Features

- **SafeSurf** is designed as an intelligent phishing detection system with the following features:
  - Real-time phishing detection using machine learning
  - Integration with Chrome extension and web interface
  - Ensemble-based classification for improved accuracy
  - DNS-based pre-filtering for faster processing
  - User-friendly output with clear security indicators
  - Scalable backend architecture

2.1.3.9 Software Requirements

- **Frontend**
  - React.js – Dynamic and responsive web interface
  - Chrome Extension – Real-time browser-based detection
- **Backend**
  - FastAPI (Python) – API handling and ML integration
- **Machine Learning**
  - Scikit-learn
  - XGBoost
  - NumPy, Pandas
- **Development Tools**
  - Jupyter Notebook
  - VS Code / Terminal
- **Deployment**
  - Docker – Containerization for consistent deployment
  - Cloud Platforms (AWS / Google Cloud)